

UNIVERSITY OF OSLO

CTaps

And gaps when supporting QUIC through
TAPS

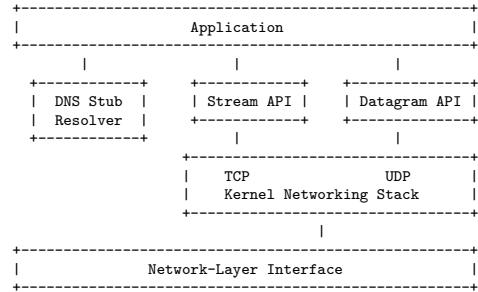
Ingebrigt Hovind

Department of Informatics

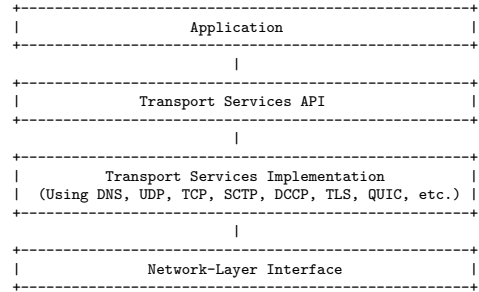
TBD, 2026



What is TAPS?



(a) Socket Programming



(b) TAPS

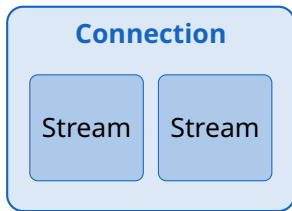
Source: RFC 9621

What is CTaps?

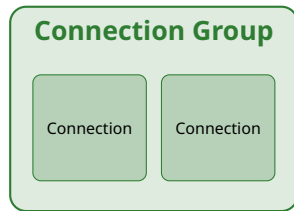
What is CTaps?

- C implementation of TAPS
- QUIC support
 - With three primary gaps

Stream closing semantics



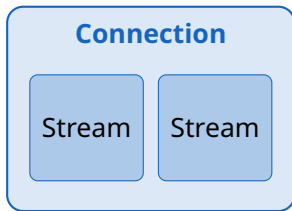
(a) QUIC



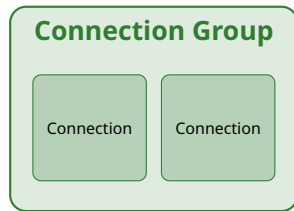
(b) TAPS

Stream closing semantics

- QUIC Connections map to Connection Groups



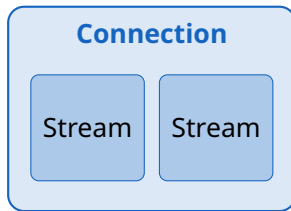
(a) QUIC



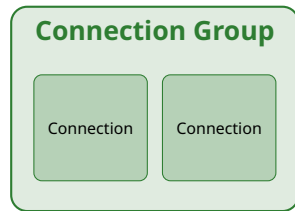
(b) TAPS

Stream closing semantics

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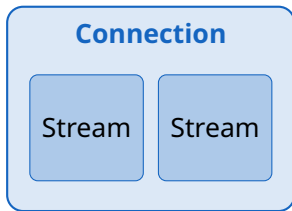
(a) QUIC



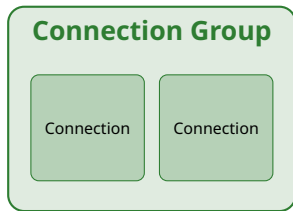
(b) TAPS

Stream closing semantics

- QUIC Connections map to Connection Groups
- Streams map to Connections
- No equivalent mapping for TCP



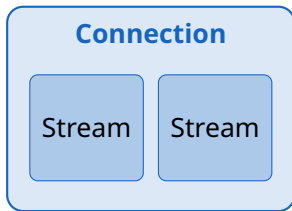
(a) QUIC



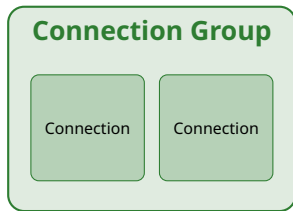
(b) TAPS

Stream closing semantics

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- On `close()`:



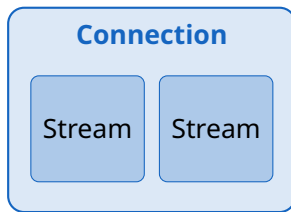
(a) QUIC



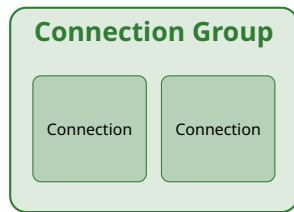
(b) TAPS

Stream closing semantics

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- No equivalent mapping for TCP
- On `close()`:
 - TCP wants to initiate graceful teardown



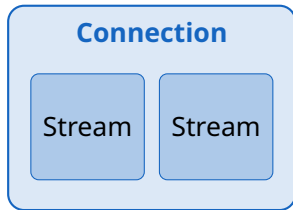
(a) QUIC



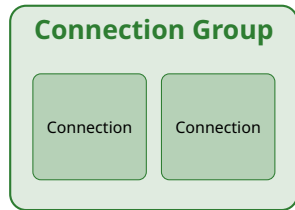
(b) TAPS

Stream closing semantics

- QUIC Connections map to Connection Groups
- Streams map to Connections
- No equivalent mapping for TCP
- On `close()`:
 - TCP wants to initiate graceful teardown
 - QUIC wants to send FIN but remain open



(a) QUIC



(b) TAPS

Aborting streams

Aborting streams

- QUIC:

Aborting streams

- QUIC:
 - STOP_SENDING
 - RESET_STREAM

Aborting streams

- QUIC:
 - STOP_SENDING
 - RESET_STREAM
- TAPS loses this directionality
 - Just a single Abort()

Stream directionality

Stream directionality

- TAPS supports directional connection establishment

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- Property is set on initial establishment

Stream directionality

- TAPS supports directional connection establishment
- Property is set on initial establishment
- Cloning and new streams happens later

Gap mitigations

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- **Directional streams and aborts**

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- Optional flags or new parameters?

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- **Directional streams and aborts**

- Update to the TAPS API
- Optional flags or new parameters?

- **Closing semantics**

- Clearly define expected behavior
- Closing Connection Group achieves same behavior accross protocols
 - ◇ Client applications utilizing TAPS must follow this advice

Conclusion

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- QUIC can be supported well in a generic API
- Some gaps do exist
 - But can be mitigated with careful design
 - Relevant for later projects
 - ◇ TAPS-based networking in C++ standard library?

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 - Relevant for later projects
 - ◇ TAPS-based networking in C++ standard library?
- Both the source code and thesis text can be found on Github:
 - <https://github.com/ikhovind/CTaps>
 - <https://github.com/ikhovind/master-thesis/releases>